

Course Title: Software Engineering

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Online Library Management System



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Software Requirement Specification (SRS)

Introduction to the SRS

The purpose of this document is to define the functional and non-functional requirements of the Online Library Management System (OLMS), which is a web-based application designed to automate and streamline the core operations of a library. This document serves as a reference for developers, designers, testers, and other stakeholders throughout the development lifecycle, ensuring a shared understanding of what the system is expected to do and how it should perform.

The scope of the system covers the complete digitization of traditional library functions such as cataloguing books, issuing and returning them, managing member accounts, calculating fines for overdue items, and generating administrative reports. By replacing manual, paper-based record-keeping with a centralized digital platform, the system aims to make library operations faster, more accurate, and accessible to both staff and members through a simple web interface.

This document is intended for software developers and system designers who will build the application, project managers overseeing its development, the quality assurance team responsible for testing it, library administrators and staff who will use it daily, and academic evaluators in case the project is developed for educational purposes. Throughout this document, certain terms are used repeatedly: SRS refers to the Software Requirement Specification itself, OLMS refers to the Online Library Management System, "Admin" refers to the library administrator or librarian, and "Member" refers to any registered user who borrows books from the library.

Functional Requirements

Functional requirements describe the specific actions that the Online Library Management System must be able to perform. Each requirement below is written with its trigger (input) and the result the system should produce (output), so that it is easy to verify later whether the built system actually satisfies it.

ID	Requirement	Input	Output
FR1	User Login	Username and Password entered by Librarian or Member	Access granted to the respective dashboard, or an "Invalid Credentials" error is shown

FR2	Search Book	Book title, author name, or subject keyword entered by the user	List of matching books displayed along with their availability status
FR3	Add / Update / Delete Book Records	Book details entered by Librarian (title, author, ISBN, category, number of copies)	Book catalogue in the database is updated and confirmation message is shown
FR4	Register / Manage Member	Member details entered by Librarian (name, member ID, contact number, course)	New member record created or existing member profile updated in the database
FR5	Issue Book	Book ID and Member ID selected by the Librarian, or requested by the Member	Book status changed to "Issued", issue date and due date generated, confirmation displayed
FR6	Return Book	Book ID of the returned book entered by the Librarian	Book status changed to "Available"; fine is calculated automatically if the return is late
FR7	Fine Calculation	Due date and actual return date of a book	Fine amount computed (if the book is late) and displayed to the Librarian and Member
FR8	View Reports	Report type selected by Librarian (issued books / overdue books / stock report)	Requested report is generated and displayed on screen
FR9	View Issue History	Member ID of the logged-in student	List of books currently issued and previously borrowed by that member is displayed

Non-Functional Requirements

Non-functional requirements do not describe what the system does, but how well it does it. These qualities are just as important as the features themselves, because a system that has all the right features but is slow, unsafe, or hard to use will still fail to satisfy its users.

ID	Category	Requirement
NFR-01	Performance	Common search and record retrieval operations should respond quickly under normal academic-library load.
NFR-02	Security	Authentication and role-based access shall prevent unauthorized administrative operations.
NFR-03	Reliability	Transactions shall be stored consistently so that issue/return records are not lost or duplicated.
NFR-04	Usability	The interface shall be simple, clear and easy for students and librarians to use.
NFR-05	Maintainability	The application shall be modular and documented so that future changes can be implemented easily.

NFR-06	Scalability	The design should support growth in the number of books, members and transactions.
NFR-07	Availability	The system should be accessible during normal library service hours with minimal downtime.
NFR-08	Data Integrity	Book availability, member and transaction data shall remain accurate after each operation.

System Modules

The major functional modules of the Online Library Management System are summarised below.

Module	Description
Authentication	Registration, login, password reset, and role validation.
Book Management	Add, update, delete, search, and availability tracking of the catalogue.
Member Management	Member registration and record maintenance.
Issue / Return	Issue, return, due-date monitoring, and fine calculation.
Reservation & Notification	Book reservations and reminders for due dates or availability.
Reports	Available, issued, overdue, and transaction summaries.

Goal of Implementation

The overarching goal of implementing this system is to automate library operations that were previously handled manually, thereby eliminating the inefficiencies and errors associated with paper-based record-keeping for tasks such as book issuing, returning, and inventory tracking. Beyond automation, the system is intended to improve accessibility for members by allowing them to search the catalogue, check the availability of books, and place reservations remotely at any time rather than being limited to library operating hours.

Another important goal is to enable more efficient resource management by giving librarians real-time visibility into inventory levels, borrowing patterns, and overdue items, which in turn helps reduce human error in areas such as fine calculation and due-date tracking that are prone to mistakes when handled manually. The system is also designed

with time efficiency in mind, aiming to shorten the time required to issue or return books and to generate operational reports compared to manual methods.

Enhancing the overall user experience is a further objective, achieved by offering both members and staff a simple, responsive, and dependable interface to interact with. The system also aims to support data-driven decision-making by providing administrators with reports and analytics that can inform decisions about procurement and the allocation of library resources. Additionally, the system is designed with scalability in mind so that it can accommodate future growth in both the size of the book collection and the number of users without requiring a complete redesign, and it places strong emphasis on security and data protection to ensure that member and institutional data remain safe throughout their storage and processing.

Assumptions and Constraints

Assumptions

- Only authorised librarians can perform issue/return and catalogue administration.
- A book can be issued only when an available copy exists and the member satisfies borrowing rules.
- A member's login credentials are assumed to be valid and protected by the user.
- Fine rules (rate, loan period) are configurable according to the institution's policy.
- The application depends on a working database and network/application environment.
- Each book title may correspond to multiple physical copies tracked collectively through a quantity field.

Constraints

- The system requires a functioning database and application environment.
- Administrative operations are restricted to authorised users only.
- Book, member, and transaction data must remain consistent at all times.
- The system is intended to operate within normal library service requirements.

- The technology stack and deployment environment may be selected during implementation.

Future Enhancements

- Reservation notifications and email/SMS/in-app alerts for due dates and overdue books.
- Barcode or RFID-based tracking for faster issue and return operations.
- Integration with digital or e-book lending services and broader digital library access.
- A dedicated mobile application for students and librarians.
- An AI-based book recommendation system to improve the user experience.
- Advanced reports, analytics, and integration with institutional systems.

Development Model Reference

The earlier phase of this project selected the Waterfall Software Development Life Cycle because the core requirements of a library system were identified as clear and relatively stable from the outset. The six sequential phases followed are: Feasibility Study (System Engineering), Requirement Analysis and Specification, Design, Coding and Unit Testing, Integration and System Testing, and Maintenance. This Phase-II document builds directly on that foundation by expanding the requirement-analysis phase into a complete, formal Software Requirement Specification.

Conclusion

This Software Requirement Specification defines the requirements and implementation goals of the Online Library Management System. The functional requirements describe the core operations such as registration, login, book search, book management, member management, issue/return, reservation, fine calculation, transaction history, and reporting. The non-functional requirements address performance, security, reliability, usability, maintainability, scalability, availability, and data integrity.

Together, these requirements provide a clear and organised foundation for the subsequent design, implementation, testing, and maintenance of the OLMS, and establish the expected users, environment, modules, assumptions, constraints, and future scope of the system.